Metrics Reference: How to Compare MicroMacro vs. Full Micro

Which metrics are correct, which are misleading, and why

Two-Scale CTMC Project

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Abstract

This document explains the full hierarchy of comparison metrics used in the project, demonstrates the mathematical mechanism behind each one, and identifies which metrics are valid under censoring and which introduce systematic bias. The central message: **conditional mean and conditional median (completed runs only) are biased whenever the two models have different completion rates. RMST is the correct primary metric.** All examples use real data from the calibrated chain-of-cliques benchmark and the CE-01 ablation experiment.

Contents

1 The Core Comparison Problem **3**

2 Analyzing Method Differences: Design Before Metrics **3**
2.1 First decide which question you are asking . . . 3
2.2 The isolation principle . . . 3
2.3 Precondition 1: report completion, then use censoring-aware estimators . . . 4
2.4 Precondition 2: horizon commensurability . . . 4
2.5 Choose a normalization independent of the effect . . . 4
2.6 Separate systematic signal from stochastic noise . . . 4
2.7 Mechanism attribution: the ablation ladder . . . 5
2.8 From attribution to a deployable parameter surrogate . . . 5
2.9 Disclose the calibration . . . 5
2.10 Include a negative control . . . 5
2.11 Checklist . . . 5

3 Metric Catalogue **6**
3.1 Completion Rate . . . 6
3.2 Conditional Mean and Median (completed runs only) . . . 6
3.3 Restricted Mean Survival Time (RMST) . . . 7
3.4 Wasserstein-1 Distance (W_1) . . . 8
3.5 Kolmogorov–Smirnov Statistic . . . 9
3.6 Per-Hop Arrival-Time Metrics: $W_1(T_i)$ and Paired Error . . . 9

4 The $I_{tot}(t)$ Plot: Phase Noise vs. Approximation Error **9**

5 Metric Hierarchy and Recommendations **10**

6 Worked Example: CE-01 Ablation **11**

7 Statistical Power and Sample Size **11**

8 Summary **12**

1 The Core Comparison Problem

We compare two simulators on the same stochastic epidemic model:

- **Full Micro**: exact Gillespie CTMC on the complete network. Ground truth.
- **MicroMacro (MM)**: two-scale approximation. Uses aggregate macro hazard $\lambda_{ij} = \beta W_{ij}\, \bar{i}_i\, \bar{s}_j$ for inter-community transfers.

The key quantity is T_{last}: the time when the epidemic first reaches the last community. In each replication the simulation runs until T_{end}; if no arrival has occurred by then, T_{last} is *censored* (unknown, only known to be $> T_{\text{end}}$).

The fundamental difficulty: **the two models typically have different completion rates** (fraction of replications where $T_{\text{last}} < T_{\text{end}}$), because MM spreads faster than Micro on average due to the Jensen effect.

2 Analyzing Method Differences: Design Before Metrics

A metric is only meaningful inside a comparison *design*. The recurring failure mode in this project is not picking the wrong formula but asking a metric to answer a question it cannot answer, or reading a number that silently mixes the effect of interest with a confounder. This section states the principles that should govern any Micro-vs-MM comparison. The metric catalogue (Section 3) is then just the toolbox used *inside* these principles, not a substitute for them.

2.1 First decide which question you are asking

Three questions look similar but demand different designs and license different conclusions:

Q1 — Existence. Does MM differ from Micro at all, beyond Monte-Carlo noise? → significance of ΔRMST (bootstrap CI) or a two-sample distributional test. Needs adequate N; nothing else.

Q2 — Magnitude and sign. In which direction, and by how much? → ΔRMST and its horizon-normalised form, plus the completion gap. Needs censoring-aware estimators and a fixed horizon.

Q3 — Mechanism / attribution. *Why* does the difference arise, and which network property controls it? → a controlled ablation that toggles one approximation at a time, plus a parameter surrogate test. Needs confounder isolation, *not* merely a large effect.

The central methodological error is answering **Q3 with a Q2 experiment**: reporting a large, uncontrolled ΔRMST and *narrating* a mechanism ("premature leakage") that the design never isolated. A magnitude is not a mechanism, and a sign that flips across scenarios (as in the β-ratio sweep) is evidence that confounders, not the named mechanism, drive the number.

2.2 The isolation principle

To attribute a difference to a cause, every *other* cause must be held fixed. Concretely:

- **Paired seeds.** Run Micro and every MM variant on the same network realisation and the same micro-layer RNG stream. The paired difference $\delta_r = T_r^{MM} - T_r^{Micro}$ then removes between-run variance, which otherwise dominates the CI. CE-01 does this; the chain-of-cliques sweep does not.

- **One factor at a time — for attribution.** To attribute error to a *mechanism*, use the ablation ladder (Section 2.7) that toggles one approximation per rung. To map *parameter dependence*, vary parameters on a grid but *record every covariate* so its effect can be regressed out; never read a single one-factor sweep as causal.
- **Commensurable conditions.** Two scenarios are comparable only if their horizon and completion regime match (Sections 2.3, 2.4). Otherwise the numbers do not live on the same axis.

2.3 Precondition 1: report completion, then use censoring-aware estimators

Report p_{Micro}, p_{MM} and their gap *before* any arrival-time number. Once the gap exceeds $\approx$ 5–10%, every completed-runs-only metric—conditional mean, conditional median, W_1, KS—is selection-biased and must *not* carry the headline (Section 3.2 shows this producing the wrong sign). Use RMST / survival curves, which include censored runs at their known lower bound. When horizons cannot be matched, fall back to a Kaplan–Meier survival comparison or re-truncate both models to a common completion level before comparing.

2.4 Precondition 2: horizon commensurability

RMST depends on T_{end}, so a raw ΔRMST is meaningful only *within* a fixed horizon. The β-ratio sweep used $T_{\text{end}} \in [2000, 20000]$; comparing its raw shifts across r compares different axes, and only the *sign* and horizon-normalised quantities survive. Rule: within a scenario report the raw shift; *across* scenarios report $\Delta\text{RMST}/T_{\text{end}}$, rel-$W_1$, or the completion-probability gap $\Delta p = p_{\text{MM}} - p_{\text{Micro}}$.

2.5 Choose a normalization independent of the effect

rel-$W_1 = W_1/\mathbb{E}[T^{Micro}]$ divides by a quantity that lives on the *same* time scale as the numerator, so it shrinks large absolute errors "for free": a longer chain inflates the denominator as fast as the numerator, and 3.5% can look reassuring by construction. Always pair it with a scale that is independent of the mean:

$$\frac{W_1}{\text{IQR}(T^{Micro})} \qquad \text{or} \qquad \frac{W_1}{\Delta T_{\text{hop}}}, \quad \Delta T_{\text{hop}} = \frac{\mathbb{E}[T_{\text{last}}]}{L-1}.$$

A W_1 that is 3.5% of the mean but 40% of a single per-hop increment is not small in any operationally relevant sense.

2.6 Separate systematic signal from stochastic noise

Between-run (phase) noise dominates unpaired comparisons and the pointwise $I_{\text{tot}}(t)$ band. Consequences:

- Prefer the *paired* δ_i for Q1/Q3: it measures only the approximation error for a shared realisation, removing phase noise.
- Read the $I_{\text{tot}}(t)$ IQR as "how variable is the epidemic", not "how biased is MM" (Section 4).
- A wide band around two nearly coincident medians is *not* evidence of a large approximation error.

2.7 Mechanism attribution: the ablation ladder

Attribution needs a ladder whose rungs each toggle exactly one approximation, so the increment in ΔRMST between adjacent rungs is charged to that one change:

Rung	Hazard × Placement	What the increment isolates
Full Micro	exact / exact	ground truth ($\Delta\text{RMST} = 0$)
Oracle E/E	exact / exact-target	numerical/synchronisation only
Oracle E/U	exact / uniform	**import placement** error
Aggregate A/B	aggregate / boundary	**rate abstraction** error
MicroMacro A/U	aggregate / uniform	placement + rate (deployed model)

CE-01 run this way attributes the bulk of the error to *rate abstraction*, with placement significant only under an exact hazard. **Caveat:** the oracle rungs read the exact edge list and node states; they are *diagnostic upper bounds*, not deployable models—a fielded MM may know only aggregate parameters, never the exact structure.

2.8 From attribution to a deployable parameter surrogate

Once the ladder localises the error to a specific approximation, the deployable question is whether a function of *known parameters*—spectral gap λ_2, mean degree, boundary-degree statistics—can reproduce the oracle correction *without* the exact structure. The correct test is not "does error correlate with λ_2 across networks" (that correlation is confounded by everything that co-varies with λ_2). It is: *does injecting the surrogate move MM up the ladder toward the oracle rung, with all other factors fixed?* To support this, a parameter sweep must record λ_2, degree, boundary statistics and the horizon per realisation, so the surrogate can be fitted and validated out-of-sample rather than fitted to the same corner cases that motivated it.

2.9 Disclose the calibration

Every current bias number assumes $\beta_{\text{macro}} = \beta$ and $W_{ij} = k$ set *by fiat*. Part of ΔRMST is then removable by tuning λ_{ij} (*calibration* error) and part is irreducible given only aggregate state (*structural* error). Report the oracle-calibrated shift as the structural floor and label every uncalibrated number as such; otherwise a reader cannot tell a property of the method from an artefact of an arbitrary choice.

2.10 Include a negative control

A credible difference analysis shows the effect *vanishing* where theory says it should. The clique–bridge–leaf topology ($n = 100$, $k = 2$) is the confounder-free control; report its W_1 and KS *with numbers*, even when null. An effect that is quantified only in corner cases (small k, slow spread, large β_{out}) and left qualitative in the clean case is not yet established.

2.11 Checklist

1. State the question: Q1, Q2, or Q3. Match the design to it.
2. Report completion first; if the gap $> 10\%$, drop completed-only metrics from the headline.
3. Fix the horizon within a comparison; normalise (and disclose T_{end}) across comparisons.
4. Add an effect-independent normalisation next to rel-W_1.
5. Use paired seeds and paired δ_i to strip phase noise.

6. For mechanism claims, show the ablation ladder, not a single big shift.
7. For parameter claims, hold covariates fixed or regress them out, and validate the surrogate out-of-sample.
8. Disclose calibration; separate structural from calibration error.
9. Include a negative control with numbers.

Pitfall	**Correct practice**
Conditional mean/median across unequal completion	Report completion; use RMST / survival (Sec. 2.3)
Raw ΔRMST compared across different T_{end}	Normalise cross-scenario; raw only within horizon (Sec. 2.4)
rel-W_1 read as "small" by itself	Add W_1/IQR or $W_1/\Delta T_{\text{hop}}$ (Sec. 2.5)
Mechanism narrated from one large shift	Ablation ladder, one toggle per rung (Sec. 2.7)
Error "explained" by correlating with λ_2	Fix covariates / regress out; validate surrogate (Sec. 2.8)
Bias numbers with $\beta_{\text{macro}} = \beta$ unlabelled	Report structural floor; mark uncalibrated (Sec. 2.9)
Clean case left qualitative	Negative control with W_1/KS numbers (Sec. 2.10)

Table 1: Mapping from common analysis pitfalls to the corresponding principle.

3 Metric Catalogue

3.1 Completion Rate

$$p_{\text{complete}} = \frac{\#\{T_{\text{last}} < T_{\text{end}}\}}{N}$$

What it measures: whether the epidemic traverses the entire chain within the simulation horizon. Affected by both the inherent stochasticity of the epidemic and by the approximation bias of MM.

Example (clique baseline, 100 reps):

$$p_{\text{Micro}} = 0.68, \quad p_{\text{MM}} = 0.82.$$

MM completes 14% more runs—direct evidence that MM spreads faster on average.

Correct use: Report it first. A large gap ($|p_{\text{MM}} - p_{\text{Micro}}| > 0.1$) is already informative and means all other completed-run metrics will be biased (see Section 3.2).

Limitation: It is a single binary threshold; it does not quantify *how much* faster or slower.

3.2 Conditional Mean and Median (completed runs only)

$$\hat{\mu}^{\text{cond}} = \frac{1}{|\mathcal{C}|} \sum_{r \in \mathcal{C}} T_r, \quad \mathcal{C} = \{r : T_r < T_{\text{end}}\}.$$

What it measures: the average (or median) arrival time *among runs that happened to complete.*

Why it is biased when completion rates differ. Suppose Jensen holds: $F^{MM}(t) \geq F^{Micro}(t)$ for all t (MM is stochastically faster, arrives earlier). The conditional median is

$$m^{Micro} = F^{-1}(0.5 \cdot p_{\text{Micro}}) = F^{-1}(0.34), \quad m^{MM} = F^{-1}(0.5 \cdot p_{\text{MM}}) = F^{-1}(0.41).$$

We compare m^{MM} to m^{Micro} but they correspond to *different percentiles* of the unconditional distribution (41st vs. 34th). Even when $F^{MM}(t) \geq F^{Micro}(t)$, the 41st percentile of Micro can easily exceed the 41st percentile of MM, making $m^{MM} > m^{Micro}$— the opposite of what we expect.

Concrete example. Clique baseline, 100 reps:

$$m^{Micro}_{\text{cond}} = 1949.4, \quad m^{MM}_{\text{cond}} = 2020.2, \quad \text{shift} = +70.8 \quad \text{(MM appears later!)}.$$

But the censoring-correct RMST shift is -75.6 (MM is earlier). The $+70.8$ is an artifact: MM's extra 14% completions are slow runs near $T_{\text{end}} = 2500$ that pull the MM conditional median upward.

Verdict: do not use conditional mean or median as the primary metric when completion rates differ. They can show the wrong sign.

3.3 Restricted Mean Survival Time (RMST)

$$\text{RMST} = \mathbb{E}[\min(T_{\text{last}}, T_{\text{end}})] = \int_0^{T_{\text{end}}} P(T_{\text{last}} > t)\, dt.$$

What it measures: the expected time until arrival, capped at T_{end}. Censored runs contribute T_{end} (their lower bound), so all runs are included without discarding information. Equivalently, RMST is the area under the survival curve $S(t) = P(T_{\text{last}} > t)$ from 0 to T_{end}.

Why it handles censoring correctly. For a censored run ($T_r > T_{\text{end}}$), we know the true arrival time is at least T_{end}. Using $\min(T_r, T_{\text{end}}) = T_{\text{end}}$ as the contribution is the correct likelihood-consistent choice for a right-censored observation.

Signed shift:

$$\Delta\text{RMST} = \text{RMST}^{MM} - \text{RMST}^{Micro}.$$

- $\Delta\text{RMST} < 0$: MM arrives *earlier* (premature propagation).
- $\Delta\text{RMST} > 0$: MM arrives *later* (over-conservative).
- $\Delta\text{RMST} = 0$: no systematic bias at the chosen horizon.

Connection to Jensen. Jensen's inequality implies $S^{Micro}(t) \geq S^{Macro}(t)$ for all t, hence $\Delta\text{RMST} \leq 0$: integrating the survival inequality gives $\text{RMST}^{MM} \leq \text{RMST}^{Micro}$. RMST is therefore the correct metric to *test* the Jensen prediction.

How RMST is computed in this project (code). From `extrascripts/benchmark_censoring_summary.p`

```
restricted_micro = np.where(np.isfinite(micro), micro, horizon)
restricted_macro = np.where(np.isfinite(macro), macro, horizon)
rmst_shift = np.mean(restricted_macro - restricted_micro)
```

Censored runs ($T_r = \text{NaN}$ or ∞) are replaced by `horizon` (T_{end}), then the shift is the mean of the pair-wise differences over all N replications. This is equivalent to the area difference between survival curves estimated by the empirical CDF.

Bootstrap confidence interval. The CI is obtained by resampling the full table of (T_r^{Micro}, T_r^{MM}) pairs (including censored rows) with replacement, 1000 times. Pairs are resampled jointly (not independently) to preserve within-run correlations if the two models share a random seed.

Concrete numbers (calibrated clique benchmark, 100 reps):

Scenario	RMST^{Micro}	RMST^{MM}	ΔRMST	95% CI
baseline $k = 5$	2089.3	2013.6	-75.6	$[-197.3, +34.8]$
$k = 10$	1326.0	1198.1	-127.9	$[-217.8, \mathbf{-39.7}]$
$k = 20$	879.5	738.7	-140.8	$[-190.6, \mathbf{-89.4}]$
$L = 15$	3342.1	3093.7	-248.4	$[-428.7, \mathbf{-51.2}]$
sparse baseline	2869.4	1640.5	-1228.9	$[-1379.4, \mathbf{-1089.6}]$

All scenarios with sufficient completion show $\Delta\text{RMST} < 0$ (bold CI = significant). The baseline $k = 5$ CI includes zero: not because Jensen is wrong, but because 100 replications are insufficient (SE ≈ 87, shift ≈ 0.87 SE).

Limitation: RMST depends on T_{end}. A longer horizon makes RMST more sensitive to slowly-completing runs. Always report the horizon alongside RMST.

3.4 Wasserstein-1 Distance ($\mathbf{W}_1$)

$$W_1(P, Q) = \int_{-\infty}^{\infty} |F_P(t) - F_Q(t)|\, dt = \int_0^1 |F_P^{-1}(u) - F_Q^{-1}(u)|\, du.$$

What it measures: the average absolute difference between the quantile functions of two distributions. Equivalently, the minimum expected distance needed to "transport" the mass of distribution P onto Q.

Relative $\mathbf{W}_1$:

$$\text{rel-}W_1 = \frac{W_1}{\mathbb{E}[T^{Micro}]}.$$

This normalises by the Micro mean, giving a dimensionless measure of distributional discrepancy.

How it is used in this project. W_1 is computed only on **completed** runs. Therefore it is subject to the same censoring bias as the conditional mean when completion rates differ. It answers: "Among runs that *both* models completed, how different are the arrival-time distributions?" This is valid for *distributional shape* comparison but does not capture the "MM completes more runs" effect.

Concrete numbers:

Scenario	W_1	rel-W_1	Note
clique baseline $k = 5$	67.0	3.5%	small, CI incl. zero
clique $k = 10$	130.1	9.8%	moderate
clique $k = 20$	140.8	16.0%	increasing with k
sparse baseline	—	42.6%	huge; 99% vs 100% complete

Limitation: W_1 on completed runs is biased when completion rates differ. For the sparse $k = 1$ scenario, Micro completes only 2% and MM completes 19%; W_1 is not reported because the two subsets are not comparable.

When to prefer W_1 over RMST:

- When both models have 100% completion (no censoring): W_1 and RMST give consistent information; W_1 is more sensitive to tail differences.
- When you want to compare distributional *shape* (not just mean): W_1 captures quantile-by-quantile differences, not only the mean.
- **Not** as the primary metric when completion rates differ.

3.5 Kolmogorov–Smirnov Statistic

$$D_{\mathrm{KS}} = \sup_t |F^{Micro}(t) - F^{MM}(t)|.$$

What it measures: the maximum vertical distance between the two CDFs. The associated p-value tests whether the two samples could have come from the same distribution.

Limitation: KS is a *qualitative* test (is there a difference?) not a *quantitative* measure (how large is the difference?). It is also computed on completed runs only. In this project it is used as a consistency check, not a primary metric.

3.6 Per-Hop Arrival-Time Metrics: $\mathbf{W_1(T_i)}$ and Paired Error

All metrics above operate on the *terminal* arrival time T_{last}. The per-hop analysis decomposes the error layer by layer.

Per-hop Wasserstein $W_1(T_i)$: Compute the W_1 distance between the distribution of arrival times T_i^{Micro} and T_i^{MM} for community $i = 0, \ldots, L-1$ (over all replications that reached community i). This answers: "Where in the chain does the distributional error peak?"

Paired cumulative error:

$$\delta_i = T_i^{MM} - T_i^{Micro} \quad \text{(same replication)}.$$

This is computed only for runs where both models reached community i (paired observation). By using the same random seed, within-run correlations are preserved and the estimator has much lower variance than between-run comparison.

Why the paired error is the most sensitive metric. Because between-run variability dominates the RMST uncertainty, the paired comparison removes it entirely: each δ_i measures only the *approximation error* for that run, not the natural stochastic variability. The cost is censoring: at the last community ($i = 9$) with 68% Micro completion, only 68 pairs exist, and the missing 32 are the *slowest* Micro runs—exactly those with the largest $|\delta_i|$.

Example (calibrated clique baseline): The median paired error at layer 7 is ≈ -200 (MM arrives 200 time units earlier); at layer 9 it reverts toward 0 due to the censoring of slow Micro runs. The IQR is ± 400—large, because each run experiences different local dynamics.

4 The $\mathbf{I_{tot}(t)}$ Plot: Phase Noise vs. Approximation Error

The total-infected count at time t:

$$I_{\mathrm{tot}}(t) = \sum_{c=0}^{L-1} I_c(t).$$

In the chain-of-cliques setup with $n = 100$ nodes per community, I_{tot} is approximately a step function: each community contributes a sigmoid step of height n.

What the IQR band on $I_{\mathrm{tot}}(t)$ actually measures. At a fixed time t, the pointwise IQR across replications reflects:

1. **Phase noise** ($\approx 90\%$ of width): natural variability in how many communities have been infected by time t. Different runs are at different "steps" of the staircase.
2. **Approximation error** ($\approx 10\%$ of width): the systematic difference between Micro and MM in how fast communities are infected.

Because phase noise dominates the IQR width, the $I_{\mathrm{tot}}(t)$ silhouette is **not a useful measure of approximation quality**. Comparing the widths of Micro and MM IQR bands tells you about natural epidemic variability, not about the bias of MM.

When the $I_{\text{tot}}(t)$ plot is useful.

- **Sanity check**: if the two medians are dramatically different (as in the sparse baseline: $\Delta I \approx 400$ at $t = 1500$), that is direct evidence of a large bias.
- **Qualitative regime identification**: the step structure shows how many communities are typically infected at each time.
- **Not for**: quantifying small-to-moderate biases (clique case), or comparing distributional shapes.

5 Metric Hierarchy and Recommendations

Metric	**Handles censoring?**	**Use when**	**Do not use when**
Completion rate	n/a	Always report first	—
RMST / ΔRMST	**Yes**	Primary metric; always valid	—
$W_1(T_{\text{last}})$	No	Both models $\approx 100\%$ complete	Completion rates differ $> 5\%$
Cond. mean shift	No	Completion rates $\approx$ equal	Completion rates differ
Cond. median shift	No	Completion rates $\approx$ equal	Completion rates differ
Paired δ_i	Partial	Sensitive error detection per hop	Last layer (censoring of slow runs)
$W_1(T_i)$ per hop	No	Error growth through chain	First/last layers with few completions
$I_{\text{tot}}(t)$	n/a	Regime identification, sanity check	Quantifying approximation error
KS statistic	No	Qualitative test only	Primary quantitative comparison

Table 2: Metric hierarchy. **Bold**: recommended primary metric.

Recommended reporting order for each scenario.

1. Completion rates p_{Micro}, p_{MM} and their difference.
2. ΔRMST with 95% bootstrap CI. Use this to test Jensen and to assess practical bias.
3. rel-$W_1(T_{\text{last}})$ *only if* both completion rates ≥ 0.9. Provides distributional shape information beyond the mean.
4. Per-hop paired error δ_i: median and IQR across replications. For identifying where in the chain the error accumulates.
5. $I_{\text{tot}}(t)$ median: for visual context only, not for quantitative conclusions.

6 Worked Example: CE-01 Ablation

CE-01 uses 100 *paired* replications (same network and seed for all variants); $T_{\text{end}} = 4000$; all five variants complete 100% of runs (no censoring). This is the ideal setting: all metrics are valid.

Primary result table.

Variant	Completion	$\mathbb{E}[T_{\text{last}}]$	ΔRMST	95% CI
Full Micro	1.00	2936.7	0	—
Oracle E/E	1.00	2944.3	+7.5	[−93.6, +120.3]
Oracle E/U	1.00	2775.5	−161.2	[−266.8, −48.8]
Aggregate A/B	1.00	2735.5	−201.2	[−302.4, −106.7]
MicroMacro (A/U)	1.00	2715.9	−220.8	[−307.9, −120.2]

Because all variants complete 100% of runs and share the same seeds, $\Delta\text{RMST} = \mathbb{E}[T^{\text{variant}}] - \mathbb{E}[T^{Micro}]$ exactly (no censoring, no between-run confounding). The conditional mean and RMST are identical here.

Why RMST still matters even at 100% completion. The RMST formulation generalises cleanly to the censored case. Using RMST throughout—even when it reduces to the plain mean—ensures the same code and the same interpretation across all experiments. This avoids introducing separate code paths that could silently compute the wrong thing when some runs are censored.

7 Statistical Power and Sample Size

A common question: how many replications are needed to detect a given ΔRMST with 80% power at the 5% significance level?

For a one-sample t-test on $\Delta_r = \min(T_r^{MM}, T_{\text{end}}) - \min(T_r^{Micro}, T_{\text{end}})$ with standard deviation σ_Δ:

$$N \approx \left(\frac{(z_{0.025} + z_{0.20})\,\sigma_\Delta}{\delta} \right)^2 = \left(\frac{2.80\,\sigma_\Delta}{\delta} \right)^2,$$

where $\delta = |\Delta\text{RMST}|$ is the target effect size.

Calibrated clique baseline ($k = 5$). From bootstrap: $\sigma_\Delta \approx 590$ (std of pairwise differences), $|\Delta\text{RMST}| = 75.6$. Required $N \approx (2.80 \times 590/75.6)^2 \approx 480$. Current 100 replications give power $\approx 25\%$—consistent with the CI including zero.

Calibrated clique $k = 10$. $\sigma_\Delta \approx 450$, $|\Delta\text{RMST}| = 127.9$. Required $N \approx (2.80{\times}450/127.9)^2 \approx 97$. 100 replications are just sufficient; CI excludes zero.

CE-01 (paired). Pairing the replications (same seed) reduces σ_Δ dramatically (between-run variance is removed). 100 paired replications are sufficient for all four contrasts with $|\Delta\text{RMST}| \geq 160$.

Practical rule of thumb: for the chain benchmark without pairing, use at least $N = 300$ replications when $|\Delta\text{RMST}| < 100$.

8 Summary

1. **RMST and** ΔRMST are the correct primary metrics. They handle censoring, connect directly to the Jensen inequality, and have well-understood statistical properties.

2. **Conditional mean/median** (completed runs only) introduces selection bias whenever completion rates differ. It can show the *wrong sign* (e.g., clique baseline: conditional median $= +70.8$ while RMST $= -75.6$).

3. $\mathbf{W_1}$ **distance** is valid for distributional shape comparison when both models have near-100% completion. Use it alongside RMST, not instead of it.

4. $I_{\text{tot}}(t)$ **pointwise plots** reflect phase noise, not approximation error. Useful for visual context (especially sparse case) but not for quantifying bias.

5. **Per-hop paired error** δ_i is the most sensitive diagnostic: it removes between-run variance by using paired seeds. Interpret the last-layer value cautiously due to censoring of slow Micro runs.

6. **100 replications** are insufficient for the clique baseline ($k = 5$) to detect the Jensen bias at 5% significance. Need $\approx$ 400–500 replications or paired seeds.